OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

Practical 1:

**Aim:** Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

**Code:**

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    else if (pid == 0)
    {
        // Child process
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
```

GNU nano 8.7.1

```c
        // Child process
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, (char *)NULL);

        // Executes only if exec fails
        perror("exec failed");
        exit(1);
    }

    else
    {
        // Parent process
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
```

**Output:**

```
yagnesh_katukala@yagnesh:~$ nano command_exec.c
yagnesh_katukala@yagnesh:~$ gcc command_exec.c -o command_exec
yagnesh_katukala@yagnesh:~$ ./command_exec
Enter a Linux command: ls

Parent Process
Parent PID: 496
Child PID: 497

Child Process
Child PID: 497
Parent PID: 496
GreeksforGeeks  command_exec.c   opareting         os_week1    os_week1.c.save  process    program    system
command_exec    opaertingsystem  opareting_system  os_week1.c  ossp_week1       process.c  program.c
Child process completed.
yagnesh_katukala@yagnesh:~$ pwd
/home/yagnesh_katukala
yagnesh_katukala@yagnesh:~$
```